

LESSON 7-3 • SESSION 1 • GUIDED NOTES**Absolute Value as Distance**

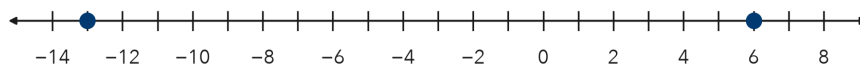
Learning target – I can use absolute value to describe how far a number is from zero. • 6.NS.C.7c

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
absolute value <i>valor absoluto</i>	How many units a number is from ____.	$ -3 = 3$ and $ 3 = 3$
distance <i>distancia</i>	How many units apart two points are. It is never ____.	from 0 to -5 is 5 units
magnitude <i>magnitud</i>	How big a number is, ignoring its ____.	-50 has magnitude 50

KEY IDEA*complete each sentence*

- Absolute value is never _____, because distance is never negative.
- Closer to 0 means a _____ absolute value.

WORKED EXAMPLE*fill in the blanks as we solve*Find $|-13|$ and $|6|$. Which number is farther from 0?**Step 1** -13 is ____ units left of 0, so $|-13| = 13$.**Step 2** 6 is 6 units right of 0, so $|6| =$ ____.**Answer** $13 > 6$, so ____ is farther from 0.**YOU TRY***use the example as a model***1.** Find each absolute value.

a. $|-21| =$ _____

b. $|4.5| =$ _____

2. Which number is closer to 0: -2.5 or 3 ? Explain with absolute value.*Show your work.***Closer to 0** _____**SUM IT UP***in your own words*

Absolute value is never negative because ____.

LESSON 7-3 • SESSION 2 • GUIDED NOTES**Absolute Value as Total Distance**

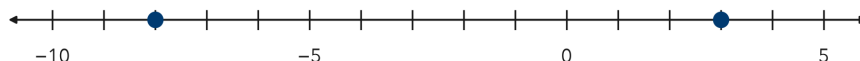
Learning target – I can use absolute value to find the total distance between a negative and a positive reading. • 6.NS.C.7c

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
absolute value <i>valor absoluto</i>	How many units a number is from ____.	$ -50 = 50$
distance <i>distancia</i>	How many units ____ two points are.	from -8 to 3 is 11 units
magnitude <i>magnitud</i>	The ____ of a number, ignoring its sign.	-50 and 50 have the same size

KEY IDEA*complete each sentence*

- If two numbers are on _____ sides of 0, add their absolute values.
- If they are on the ____ side of 0, subtract the smaller absolute value from the larger.

WORKED EXAMPLE*fill in the blanks as we solve*Each unit is 1 block. How far is it from -8 to 3 ?**Step 1** 0 is between them, so they are on _____ sides.**Step 2** $|-8| = 8$ and $|3| = \underline{\hspace{1cm}}$.**Answer** $8 + 3 = \underline{\hspace{1cm}}$ blocks**YOU TRY***use the example as a model*

1. Find the distance between each pair.

a. -5 and 9 _____**b.** -12 and -4 _____

2. A submarine rises from
- -30
- m to
- -12
- m. How far does it rise?

*Show your work.***Answer** _____ m**SUM IT UP***in your own words*

To find the distance between a negative and a positive number, I ____ because ____.
